

Disproof of TLMC Conjecture 00000000602

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Conjecture 1 (00000000602). Let V_K denote the Jones polynomial and let ω be a primitive fourth root of unity. For every alternating knot K :

1. $|V_K(\omega)|^2$ divides $\det(K)^2$, and
2. the quotient $\det(K)^2/|V_K(\omega)|^2$ equals $|V_K(i)|^2$.

Verdict

FALSE. The conjecture already fails at the trefoil, the smallest alternating knot.

The attack: $\omega = \pm i$

The primitive fourth roots of unity are exactly $\omega = \pm i$, whose complex conjugates are their negatives. Since the Jones polynomial of a knot has real coefficients, $V_K(\bar{z}) = \overline{V_K(z)}$, and therefore

$$|V_K(\omega)|^2 = |V_K(\pm i)|^2 = |V_K(i)|^2.$$

So clause (2) collapses to the requirement

$$\det(K)^2 = |V_K(\omega)|^2 \cdot |V_K(i)|^2 = |V_K(i)|^4. \quad (1)$$

Counterexample: $K = 3_1$

Take the right-handed trefoil 3_1 , which is alternating. In the standard normalization $V(\text{unknot}) = 1$,

$$V_{3_1}(t) = t + t^3 - t^4.$$

This was recomputed from scratch (independent of tables) via the Kauffman bracket state-sum on the PD code $X(1, 4, 2, 5)$, $X(3, 6, 4, 1)$, $X(5, 2, 6, 3)$, which yields $\langle 3_1 \rangle = A^{-7} - A^{-3} - A^5$ and, with $V(t) = (-A)^{-3w} \langle K \rangle|_{A=t^{-1/4}}$, $w = 3$, exactly $t + t^3 - t^4$; the normalization check $V(1) = 1$ passes.

Evaluating exactly on Gaussian integers:

$$V_{3_1}(i) = i + i^3 - i^4 = i - i - 1 = -1, \quad V_{3_1}(-i) = -1,$$

so

$$|V_{3_1}(\omega)|^2 = |V_{3_1}(i)|^2 = 1 \quad \text{for both } \omega = \pm i.$$

The determinant is $\det(3_1) = |\Delta(-1)| = 3$ with Alexander polynomial $\Delta(t) = t^{-1} - 1 + t$, hence $\det(3_1)^2 = 9$. Now:

- Clause (1) happens to hold: $1 \mid 9$.
- Clause (2) fails: the quotient is $9/1 = 9$, but $|V_{3_1}(i)|^2 = 1$, and $9 \neq 1$.

Equivalently, the collapsed requirement (1) reads $9 = 1$, which is absurd. Hence the conjecture is false.

Further witnesses

The same exact evaluation applied to the published Jones polynomials of other small alternating knots (all recomputed in the accompanying `reproduce.py`) shows the failure is not an accident of the trefoil: in every case $|V_K(i)|^2 = 1$ while $\det(K)^2 > 1$.

K	$\det(K)$	$\det(K)^2$	$ V_K(i) ^2$	quotient vs. $ V_K(i) ^2$
3_1	3	9	1	$9 \neq 1$
4_1	5	25	1	$25 \neq 1$
5_1	5	25	1	$25 \neq 1$
5_2	7	49	1	$49 \neq 1$
7_1	7	49	1	$49 \neq 1$

Table 1: All five alternating knots satisfy clause (1) trivially ($|V_K(\omega)|^2 = 1$) but violate clause (2).

Boundary of the disproof

- Only the *quotient identity* (clause 2) is refuted; the divisibility clause (clause 1) is untouched by this counterexample.
- Since $|V_K(i)|^2 = 1$ for all alternating knots tested, the identity clause would force $\det(K) = 1$; restricting the conjecture to high-crossing or high-determinant knots cannot repair it.
- The numbers $|V(\pm i)|^2 = 1$ are stable under mirroring ($V_{\bar{K}}(t) = V_K(1/t)$) and conjugation, so the chirality convention is irrelevant to the disproof.

Machine verification and reproduction

The directory `lean4/` contains a core-Lean 4 (no Mathlib) package that re-derives the counterexample from a from-scratch Gaussian-integer encoding: $V(\pm i) = (-1, 0)$, $|V(\omega)|^2 = 1$, $\det(3_1)^2 = 9$, and $\neg(9 = 1 \cdot 1)$. All proofs are `rfl/decide` kernel computations; there is no `sorry`, and `Check.lean` audits that every theorem *does not depend on any axioms* (not even `propext`, `Quot.sound` or `Class.choice`).

To reproduce all numbers:

```
python3 reproduce.py          # exact state-sum + Gaussian evaluation
cd lean4 && lake build && lake env lean Check.lean  # Lean proof + axiom audit
```